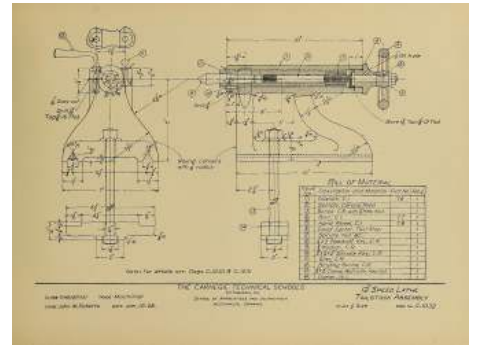
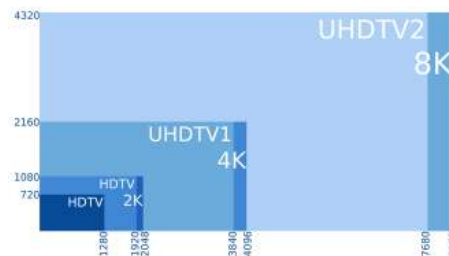




TOPIC OVERVIEW

Topic B — Dilations and Similarity

Scale Factor, Similar Figures, and Area & Perimeter



This overview guide contains...

The topic calendar — every objective and scheduled assessment across this part of the unit

Vocabulary — the terms you must build into your artifacts

Check Your Understanding problem sets — an extra tool to monitor your progress. Completing all questions is not required, but a guide kept up with, completed in full, and turned in at the end of the quarter raises your Standards Mastery Check score by **5%**. Answer keys for every question are posted on the class website.

Since this guide contains so much foundational information vital to success in this course, it is **essential** to keep up with it. It's THE **most important document** needed for success.

Topic Calendar

Monday, August 31st – Friday, September 11th

ARTIFACT 5 · Dilations and Similar Figures

WEEK 4 AUGUST 31–SEPTEMBER 4

BLOCK 1

MON–TUE

AUG 31–SEP 1

I can identify the vertices of an image transformed in a way that does not preserve congruence, when given the vertices or graph of the pre-image. **8.10B**

I can determine the proportional relationship of corresponding sides of similar figures, where the figures have a common side, even with a rotated or reflected image. **8.3A**

I can write a statement about the relationship of the side lengths of a dilated image given the graph of the pre-image. **8.3B**

ARTIFACT 6 · Algebraic Rules of Dilations

ARTIFACT 7 · Area and Perimeter of Dilations

BLOCK 2

WED–THU, SEP 2–3

I can determine the algebraic representation of a dilation when given coordinates and graphs of the pre-image and image, including only being given a graph of the pre-image and the ordered pair of a corresponding point on the graph of the image. **8.3C**

I can identify the ordered pair on the graph of a dilation when given the scale factor as a variable. **8.3C**

I can determine the ratios of area and perimeter of the pre-image and dilated image, including through expressions. **8.10D**

Feedback & Revision Studio · Performance Tasks 1.1 and 1.2

Friday, Sep 4 · also this day: Q1 Mathia Mid-Quarter Check

WEEK 5 SEPTEMBER 7–11

no school Monday

BLOCK 3

TUE–WED, SEP 8–9

Block 2, objectives 1 and 2, continued. **8.3C**

Performance Task 1.3 · Representing Scale Factor Algebraically

in class Tue–Wed, Sep 8–9 · revisions due Friday, Sep 18

BLOCK 4

THU–FRI, SEP 10–11

Block 2, objectives 1 and 2, continued. **8.3C**

Project Write-Up · Improve Dallas

due Friday, Sep 11

AFTER THE TOPIC

Unit 1 Mid-Unit Assessment · all TEKS to date (8.10A–D, 8.3A–C)

Friday, Sep 18 · revisions due Friday, Oct 2

Vocabulary

All of these terms must be eventually added to at least one artifact (Artifact 5, 6, and/or 7!)

pre-image	the original figure (e.g. $ABCD$)
image	the figure after the transformation, named with prime notation (e.g. $A'B'C'D'$)
corresponding sides/angles/vertices	the matching parts of the pre-image and image
congruent/congruence	same size <i>and</i> shape; a dilation with a scale factor other than 1 is not congruent
rigid motion	a move that keeps the same size and shape (translation, reflection, rotation); a dilation is not a rigid motion
dilation	a transformation that resizes a figure by a scale factor from a center point; it keeps the same shape but changes the size
scale factor	the number every distance from the center is multiplied by; from the origin, $(x, y) \rightarrow (kx, ky)$ – e.g. $k = 2$ doubles, $k = \frac{1}{2}$ halves
center of dilation	the fixed point a figure is resized from (here, usually the origin)
origin	the center of the coordinate plane, point $(0, 0)$; the usual center of dilation
enlargement	a dilation with a scale factor greater than 1 (the image is larger)
reduction	a dilation with a scale factor between 0 and 1 (the image is smaller)
similar figures	figures with the same shape but not necessarily the same size: corresponding angles are equal and corresponding sides are proportional (symbol $\sim$)
similarity	the relationship between two similar figures
proportional	having a <u>constant ratio</u> ; in a dilation every side of the image is the same number of times its matching side

$0 < k < 1$

Similar figures can be congruent, but aren't required

Corresponding side lengths + coordinates must be proportional!

or equivalent ratio.

ratio	a comparison of two quantities by division, e.g. $A'B' : AB = 2 : 1$
scale model	a real-world enlargement or reduction that keeps the same proportions as the original (e.g. a museum's blue-whale model)
perimeter	the distance around a figure; a dilation by scale factor k multiplies the perimeter by k
area	the amount of surface a figure covers; a dilation by scale factor k multiplies the area by k^2
algebraic representation (rule) of a dilation	the dilation written as mapping notation from the origin: $(x, y) \rightarrow (kx, ky)$
variable scale factor (k)	using a letter k in place of a number so one rule works for any size
expression	a math phrase that may use a variable (e.g. $4x$, k^2); area and perimeter ratios can be written as expressions
vertex/vertices	the corner point(s) of a figure
coordinate plane	a four-quadrant grid with two axes (x-values and y-values) on which transformations of figures are represented

Check Your Understanding

Topic B – Dilations and Similarity

Scale Factor, Similar Figures, and Area & Perimeter

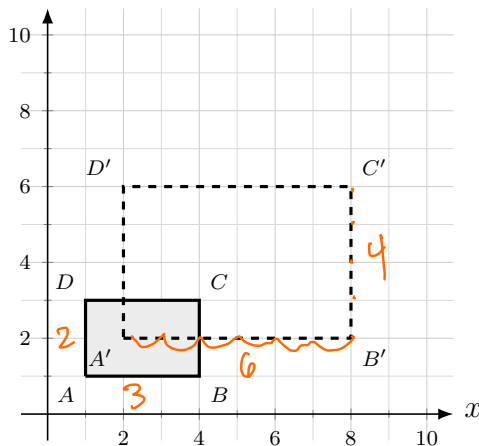
Questions 1-12 cover Artifact 5 (Dilations and Similar Figures)

1. In your own words, describe what a **dilation** does to a figure. Then explain why a dilation by a scale factor of 2 does *not* produce a congruent image.

A dilation changes the size of a figure by multiplying coordinate values / side lengths by a scale factor (k). Since this enlarges or reduces side lengths / angles by a factor of 2

2. Rectangle $ABCD$ is dilated from the origin to form rectangle $A'B'C'D'$, as shown.

(doubles), the image is not congruent to the pre-image.



- (a) Write the ratio of side $A'B'$ to side AB .

$A'B' : AB$ - or - $\frac{A'B'}{AB}$ ← preferred

- (b) Is the image congruent to the pre-image? Explain how you know.

We can compare side lengths & area between image & pre-image (both must be the same to be congruent).

Side lengths

$\overline{AB} = 3$ units ; $\overline{A'B'} = 6$ units

Area

$ABCD = 2 \cdot 3 = 6$ sq. units
 $A'B'C'D' = 4 \cdot 6 = 24$ sq. units

3. A triangle has side lengths 3, 4, and 5. It is dilated by a scale factor of 3. Write one statement describing how every side length of the image compares to the original, and give the three new side lengths.

* Dilations require we multiply all by the scale factor ($k=3$).

* All new side lengths are triple the length of the original.

pre-image side lengths		image side lengths
3	$\times 3 =$	9
4	$\times 3 =$	12
5	$\times 3 =$	15

4. A figure is dilated by a scale factor of 2 from the origin. Which statement is true?

(A) The image is congruent to the original figure.

(B) Each side length of the image is half the matching side length of the original.

(C) The corresponding angle measures stay the same, and each side length doubles.

(D) The image has the same perimeter as the original figure.

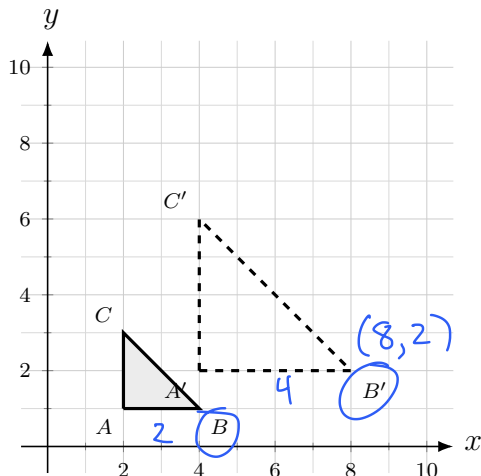
→ does not preserve congruence $k=2$

is multiplied by 2 → gets bigger (enlargement)

$\times 2 = \text{doubling}$ ✓

perimeter is also $\times 2$!

5. Triangle ABC is dilated from the origin to form triangle $A'B'C'$, as shown.



(a) Which vertex of the image corresponds to vertex B , and what are its coordinates? Vertex B corresponds with vertex B' .

Located at $B'(8,2)$

(b) Explain how you know this transformation does not preserve congruence. I can compare side lengths or area. Let's do both

just in case:

2 corresponding side lengths
 $\overline{AB} = 2$ units
 $\overline{A'B'} = 4$ units

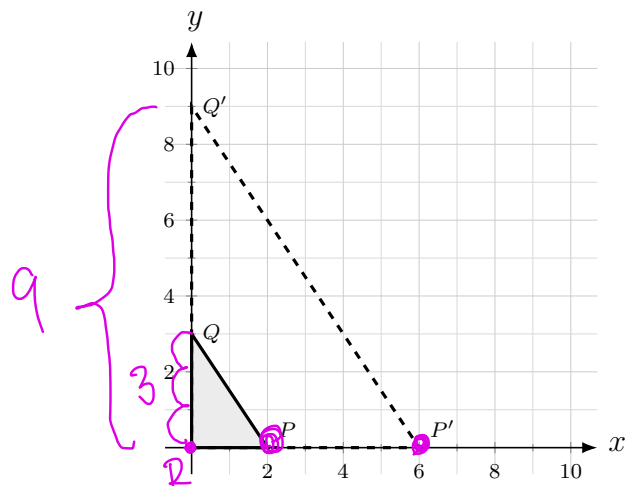
not congruent

Both areas

$A(\triangle ABC) = \frac{1}{2} \cdot 2 \cdot 2 = 2$
 $A(\triangle A'B'C') = \frac{1}{2} \cdot 4 \cdot 4 = 8$

-or- We know dilations > 1 do not preserve congruence

6. Two similar triangles share the same vertex at the origin, as shown. The smaller triangle is the pre-image and the larger triangle is its dilation.



(a) Write the ratio of a side of the larger triangle to its corresponding side of the smaller triangle.

$\frac{Q'P'}{QP}$ - or - $\frac{Q'R'}{QR}$ - or - $\frac{P'R'}{PR}$

(b) What is the scale factor of the dilation?

Take the ratio of $\frac{\text{image}}{\text{pre-image}}$ or $\frac{\text{new}}{\text{original}}$ & divide:

w/ side lengths $\frac{Q'R'}{QR} = \frac{9}{3} = 3$ w/ coordinates $\frac{P'}{P} = \frac{6}{2} = 3$

7. Two figures are **similar**. Which statement *must* be true?

(A) The two figures are the same size. can be, not must

(B) The corresponding angles are equal and the corresponding sides are proportional.

(C) The corresponding sides are equal in length. can be, not must

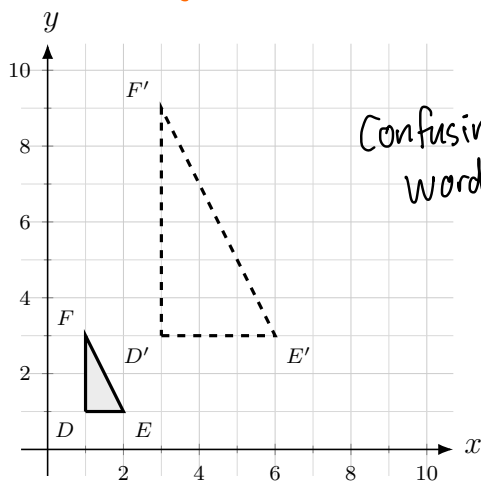
(D) The two figures have the same perimeter.

every corresponding side ratio equivalent

$\frac{6}{2} = 3$
 $k=3$

Scale factor $\Rightarrow k = ?$

- * 8. Triangle DEF is dilated from the origin to form triangle $D'E'F'$, as shown.



Confusing wording

Write one statement describing how every side length of $D'E'F'$ compares to triangle DEF , and state what that makes the two figures.

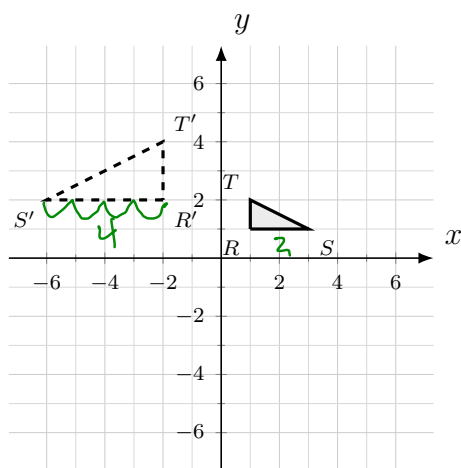
$$\frac{D'F'}{DF} = \frac{D'E'}{DE} = \frac{E'F'}{EF}$$

$k = \frac{1}{2}$

9. A figure is dilated from the origin by a scale factor of $\frac{1}{2}$. Is this an **enlargement** or a **reduction**? Describe what happens to the figure and to each of its side lengths.

Since $0 < k < 1$, (essentially multiplying by a scale factor less than 1) this is a reduction. Multiplying any side length or coordinate by a number less than 1 decreases the value.

10. Triangle RST (the pre-image) is reflected across the y -axis and dilated to form the similar triangle $R'S'T'$, as shown. Because the image is flipped, match the sides by correspondence, not by position.



not just dilated about $\rightarrow$ Don't rely on origin coordinates here $\rightarrow$ side lengths

- (a) Write the ratio of side $R'S'$ to its corresponding side RS .

$$\frac{R'S'}{RS} = \frac{4}{2} = 2$$

- (b) Are the two triangles similar, congruent, or both? Explain. They are not congruent, as both image & pre-image have different corresponding side lengths. But since they're proportional (equivalent),

11. Determine whether each transformation produces a **congruent** image or a **similar but not congruent** image. Place a check in the correct column for each row.

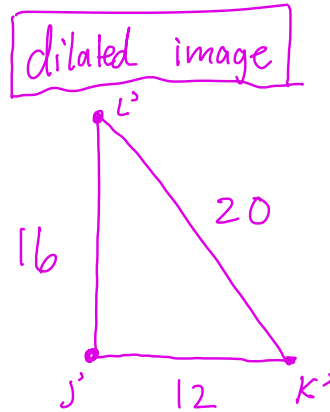
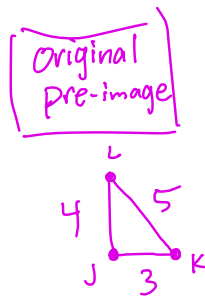
	Congruent image	Similar but not congruent
A dilation by a scale factor of 2 *enlargement	☐	☒
A reflection across the x -axis *No change in size or shape	☒	☐
A dilation by a scale factor of $\frac{1}{2}$ *reduction	☐	☒
A translation 3 units right and 1 unit up	☒	☐

* No change in size or shape

12. A student claims that two similar figures must be congruent. Decide whether this is correct, and justify with a labeled dilation example that refers to corresponding angles, corresponding sides, and the scale factor.

While similar figures can happen to be congruent, and while shape must stay the same (angles, proportional side lengths), size (area/side lengths)

can differ & still be similar, as shown below:



Corresponding side lengths are proportional

$$\frac{J'K'}{JK} = \frac{12}{3} = 4 \checkmark$$

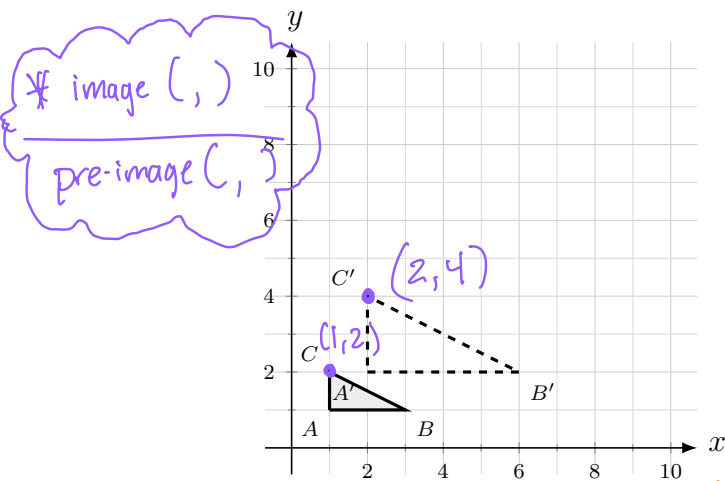
$$K = 4$$

$$\frac{K'L'}{KL} = \frac{20}{5} = 4 \checkmark$$

$$\frac{J'L'}{JL} = \frac{16}{4} = 4 \checkmark$$

Questions 13-24 cover Artifact 6 (Algebraic Rules of Dilations)

13. Triangle ABC is dilated from the origin to form triangle $A'B'C'$, as shown. Write the algebraic rule for the dilation on the line beneath the prompt.



$$\frac{C'(2,4)}{C(1,2)} \quad \frac{4}{2} = 2 \quad k=2$$

Write the algebraic rule $(x,y) \rightarrow (,)$ for this dilation.

$$(x,y) \rightarrow (2x, 2y)$$

$$(x,y) \rightarrow (kx, ky)$$

Dilation algebraic rule, always: $(x,y) \rightarrow (kx, ky)$

14. Write the algebraic rule for each dilation from the origin. Write your rule in the parentheses.

- (a) scale factor of 2

$$(x,y) \rightarrow (2x, 2y)$$

- (b) scale factor of 3

$$(x,y) \rightarrow (3x, 3y)$$

- (c) scale factor of $\frac{1}{2}$

$$(x,y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$$

- (d) scale factor of 5

$$(x,y) \rightarrow (5x, 5y)$$

15. A figure is dilated from the origin by a scale factor of 3. Which rule represents this dilation?

~~(A) $(x,y) \rightarrow (x+3, y+3)$~~

→ (B) $(x,y) \rightarrow (3x, 3y)$

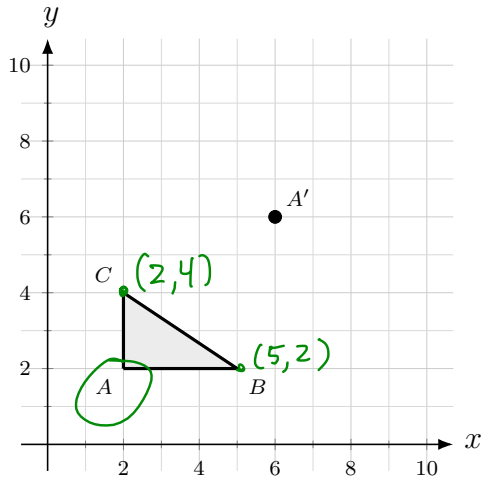
~~(C) $(x,y) \rightarrow (3x, y)$~~

~~(D) $(x,y) \rightarrow (x, 3y)$~~

$$k=3$$

$$(x,y) \rightarrow (3x, 3y)$$

16. Triangle ABC has vertices $A(2, 2)$, $B(5, 2)$, and $C(2, 4)$. After a dilation from the origin, only one image point was recorded: $A'(6, 6)$, shown as the dot.



$$k = \frac{A'}{A} = \frac{(6, 6)}{(2, 2)} = \frac{6}{2} = 3$$

(a) Use A and A' to find the scale factor, then write the dilation rule.

$$(x, y) \rightarrow (3x, 3y)$$

(b) Use your rule to find the coordinates of B' and C' . $\ast$ Multiply all coordinates by $k=3$

pre-image		image
$B(5, 2)$	$\xrightarrow{\times 3}$	$B'(15, 6)$
$C(2, 4)$	$\xrightarrow{\times 3}$	$C'(6, 12)$

17. A figure is dilated from the origin by the rule $(x, y) \rightarrow (2x, 2y)$. Find the image of each point: $D(3, 5)$ and $E(4, 1)$.

pre-image		image
$D(3, 5)$	$\xrightarrow{\times 2}$	$D'(6, 10)$
$E(4, 1)$	$\xrightarrow{\times 2}$	$E'(8, 2)$

18. A figure is dilated from the origin by a scale factor of k . Write the coordinates of the image of point $P(5, 2)$ in terms of k .

The rule: $(x, y) \rightarrow (k \cdot x, k \cdot y)$
 $\rightarrow$ We don't know k - keep it there, replace x - & y -values

$$P'(5k, 2k)$$

19. A dilation from the origin maps point $Q(3, 4)$ to $Q'(12, 16)$. What is the scale factor, and what is the algebraic rule for the dilation?

pre-image		image
$Q(3, 4)$	$\xrightarrow{k=?}$	$Q'(12, 16)$

$$k = \frac{12}{3} = 4 \checkmark \quad \text{or try } y\text{-values} - \quad k = \frac{16}{4} = 4 \checkmark$$

$\ast$ Dilations apply to both coordinates evenly.

20. A part is dilated from the origin by a scale factor of k . A client needs the image of point $B(5, 2)$ to land at $(15, 6)$. What scale factor k produces this, and how do you know?

pre-image		image
$B(5, 2)$	$\xrightarrow{k=?}$	$B'(15, 6)$

$$k = \frac{6}{2} = 3$$

$\ast$ I know because I divided corresponding coordinates $\left(\frac{\text{new}}{\text{original}} \text{ or } \frac{\text{image}}{\text{pre-image}}\right)$ to get the scale factor.

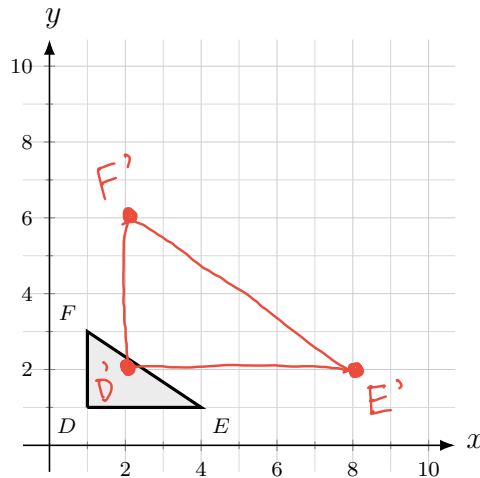
21. Triangle DEF with vertices $D(1, 1)$, $E(4, 1)$, and $F(1, 3)$ is dilated from the origin by a scale factor of 2. Plot the image $D'E'F'$ on the grid below, and label the coordinates of each vertex.

Pre-image | image'

$D(1, 1) \xrightarrow{\times 2} D'(2, 2)$

$E(4, 1) \xrightarrow{\times 2} E'(8, 2)$

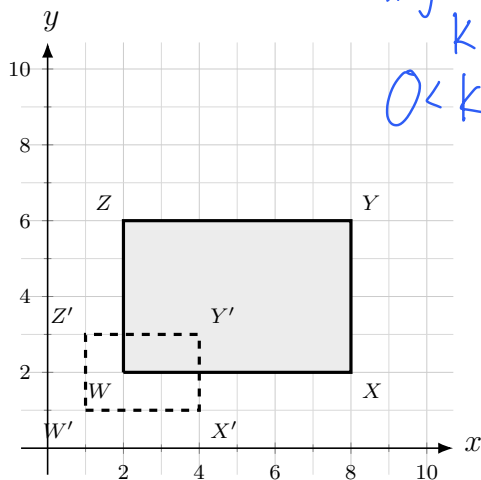
$F(1, 3) \xrightarrow{\times 2} F'(2, 6)$



22. Determine whether each rule represents a **dilation**, a **translation**, or a **reflection**. Place a check in the correct column for each row.

"2 times x, 2 times y"	Dilation	Translation	Reflection
$(x, y) \rightarrow (2x, 2y)$ *multiplication $\rightarrow$ dilation	☒	☐	☐
$(x, y) \rightarrow (x + 2, y)$ *translation $\rightarrow$ translation	☐	☒	☐
$(x, y) \rightarrow (-x, y)$ *reflection across y-axis	☐	☐	☒
$(x, y) \rightarrow (3x, 3y)$ another dilation	☒	☐	☐

23. Rectangle $WXYZ$ (the pre-image) is dilated from the origin to form the smaller rectangle $W'X'Y'Z'$, as shown.



*gets smaller $\rightarrow$ reduction
 k must be between 0 & 1
 $0 < k < 1$

- (a) Write the algebraic rule for this dilation.

$$(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y \right)$$

- (b) Is this dilation an enlargement or a reduction? How does the rule tell you?

Since the rule contains a multiplier (scale factor) between 0 & 1 ($0 < k < 1$), we know side lengths + area will get smaller

*Compare corresponding coordinates

$Y(8, 6) \xrightarrow{k=?} Y'(4, 3)$

*going backwards

$6 \cdot \frac{1}{2} = 3$

$8 \cdot \frac{1}{2} = 4$

24. Explain why writing a dilation as a rule with a variable scale factor, $(x, y) \rightarrow (kx, ky)$, is more useful than a rule for one fixed size. State what stays the same about a figure's shape for any value of k , using corresponding sides and proportionality.

The multiplication rule in a dilation doesn't specify exact locations because it can still apply the same proportional multiplicative relationship on any coordinate value while still producing proportional sides.

Computer designers & game developers can rely on a single dilation rule to apply to various graphics at various points on a coordinate grid (creating an individual rule for each graphic would be a lot of work!).

Questions 25-35 cover Artifact 7 (Area and Perimeter of Dilations)

25. A rectangle is 3 units wide and 2 units tall. It is dilated from the origin by a scale factor of 2.

(a) Find the perimeter of the original rectangle and of the dilated image, then write the ratio (image:original).

$P(\text{original}) = 3 + 2 + 3 + 2 = 10$
 $P(\text{dilated image}) = 6 + 4 + 6 + 4 = 20 \text{ units}$
 image:original $\rightarrow 20:10 \rightarrow 2:1$

(b) Find the area of each rectangle and write that ratio (image:original).

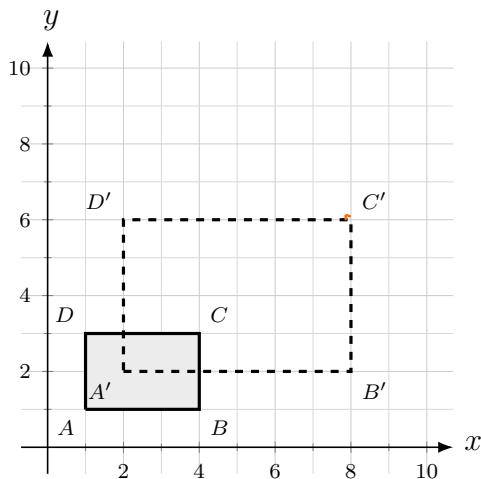
$A(\text{original}) = 3 \cdot 2 = 6$
 $A(\text{dilated}) = 6 \cdot 4 = 24$
 image:original $\rightarrow 24:6 = 4:1$

26. A figure is dilated from the origin by a scale factor of 3. The area of the image is how many times the area of the original figure?

- (A) 3 times
 (B) 6 times
 (C) 9 times
 (D) 12 times
- $K = 3$
 $\text{times } 3^2 = 9$
 Perimeter: $K:1$
 Area: $K \cdot K:1 \leftarrow \text{multiply by } K^2 \text{ to get the area}$

* 27. Rectangle $ABCD$ is dilated from the origin by a scale factor of 2 to form $A'B'C'D'$, as shown.

Same as 25...



Find the perimeter ratio and the area ratio ($A'B'C'D' : ABCD$). Show the measures you used.

28. Complete the table for a dilation by scale factor k . Write each ratio as image:original. The bottom row uses the variable k .

Scale factor	Perimeter ratio	Area ratio
$k = 2$	$2:1$ (just the scale factor)	$2 \cdot 2 \rightarrow 4:1$
$k = 3$	$3:1$	$3 \cdot 3 \rightarrow 9:1$
$k = 5$	$5:1$	$5 \cdot 5 \rightarrow 25:1$
k (any)	$K:1$	$K \cdot K \rightarrow K^2:1$

Just multiply the scale factor twice!

29. Moving an image from 1080p to a 4K screen dilates it by a scale factor of 2. Explain why the 4K screen has *four* times as many pixels, using how area changes under a dilation.

When calculating area of a dilated figure, since all side lengths multiply by 2, and you have to use all side lengths to calculate area, you are ² essentially multiplying by the scale factor twice: $k \cdot k$ (k^2).

30. A figure is dilated from the origin by a scale factor of $\frac{1}{2}$. What happens to its area? Write the area ratio (image:original) and explain your reasoning.

$$\begin{aligned} \text{Area ratio (image:original)} &= k^2 : 1 \\ &= \frac{1}{2} \cdot \frac{1}{2} : 1 = \frac{1}{4} : 1 \end{aligned}$$

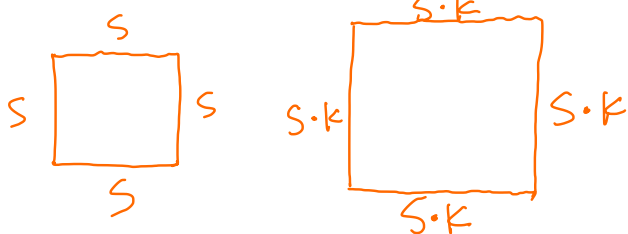
$k = \frac{1}{2}$

31. The area of a dilated image is 16 times the area of the original figure. What scale factor was used? Explain how you found it.

Using the area ratio $\rightarrow k^2 : 1$
 $16 : 1$

So k^2 must equal 16
 $k \cdot k = 16$
 What exact number, times itself, equals 16? 4
 $k = 4$

32. A square has a side length of s . It is dilated from the origin by a scale factor of k . Write an expression for the **perimeter** of the image and an expression for the **area** of the image, each in terms of s and k .



$$\begin{aligned} P(\text{image}) &= 4 \cdot s \cdot k \\ A(\text{image}) &= (s \cdot k)^2 = s^2 k^2 \end{aligned}$$

33. For a dilation by scale factor k , determine whether each quantity is multiplied by k or by k^2 . Place a check in the correct column for each row.

	Multiplied by k	Multiplied by k^2
The perimeter of the figure just $\cdot k$	☒	☐
The area of the figure $\cdot k^2$	☐	☒
The length of one side (perimeter $\rightarrow \cdot k$)	☒	☐
The <u>distance</u> from the origin to a vertex	☒	☐

↳ Still talking about length, not area

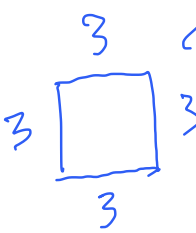
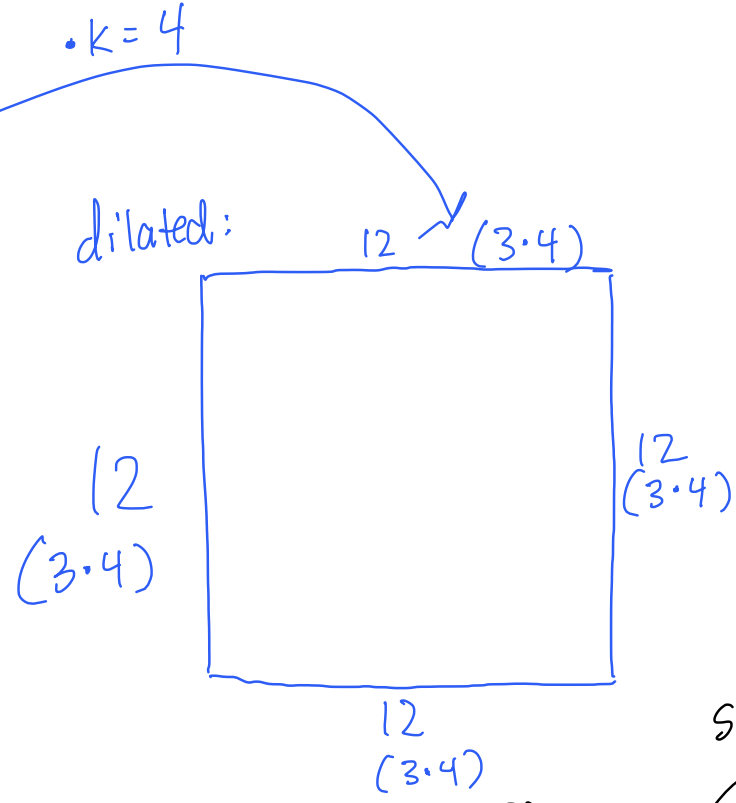
34. A designer builds a small reference tile and then a version of it dilated by a scale factor of 4. Without computing exact numbers, predict how many times larger the *perimeter* of the big tile is and how many times larger its *area* is. Justify each prediction.

For perimeter, length changes directly by how many times individual side lengths change ($\cdot k$), so it will be 4 times larger. For area, it multiplies by the scale factor times itself $\rightarrow k^2 \rightarrow$ so it will be $4 \cdot 4 = 16$ times the size.

35. Explain why dilating a figure by a scale factor of k multiplies its **perimeter** by k but its **area** by k^2 . Support your answer with a worked numerical example.

This is because area incorporates the multiplicative Scale factor twice ($k \cdot k$). Since perimeter only adds side lengths, the only multiplication is with the original scale factor, which multiplies by length once, not twice as in the area formula.

Example:

original:  dilated: 

Perimeter

$$P(\text{original}) = 3 + 3 + 3 + 3 = 12$$

$$P(\text{dilated}) = 12 + 12 + 12 + 12 = 48$$

Perimeter changed $\times 4$, just like the scale factor

Area

$$A(\text{original}) = 3 \cdot 3 = 9$$

$$A(\text{dilated}) = (3 \cdot 4)(3 \cdot 4) = 12 \cdot 12 = 144$$

Scale factor multiplied twice

$$144 \div 9 = 16$$

Area changed by $4 \cdot 4$ (16 times).